Machine Learning & Pattern Recognition

The ML Landscape

Understanding how computers learn from data.

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Spam Email

Unwanted and often harmful emails

From: winner@prize2024.com
To: you@example.com
Subject: **YOU WON A FREE IPHONE!!!**

Congratulations!!! 🎉
You have won a **FREE iPhone 15!**
Click here now to claim your prize:

👉 www.free-iphone-now.com

Hurry! This offer is valid for a short time only!

Best regards,
Prize Team



Why it's Spam?

- Suspicious sender address
- Too good to be true offer
- Urgent language (Hurry!)
- Contains a link to an unknown website
- Unsolicited (you didn't request it)

Spam

Non-Spam Email (Ham)

Legitimate and useful emails

From: noreply@university.edu
To: you@example.com
Subject: **Your Course Schedule for Fall 2025**

Mansoura University

Dear Student,
Your course schedule for Fall 2025 is now available on the student portal. Please log in to view your registered courses.

View Your Schedule

If you have any questions, feel free to contact the IT support team.

Best regards,
Registrar Office
Mansoura University

Why it's Not Spam?

- Trusted and recognizable sender
- Relevant and important information
- No suspicious links
- Professional and clear language
- Expected email (you are a student)

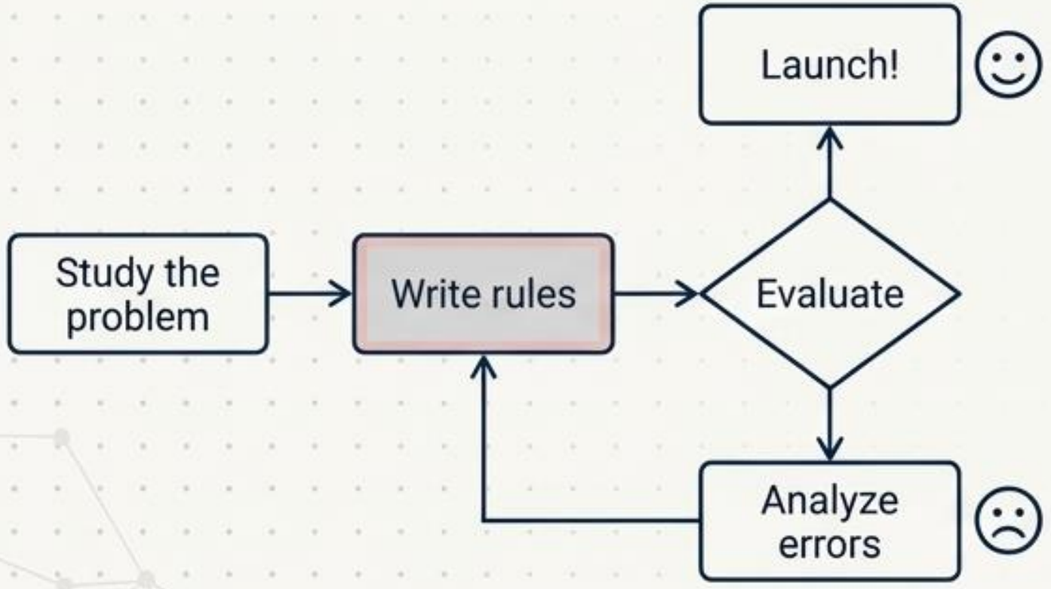
Not Spam (Ham)

Why do we need Machine Learning?

The Spam Filter Problem: Traditional vs. ML Approach

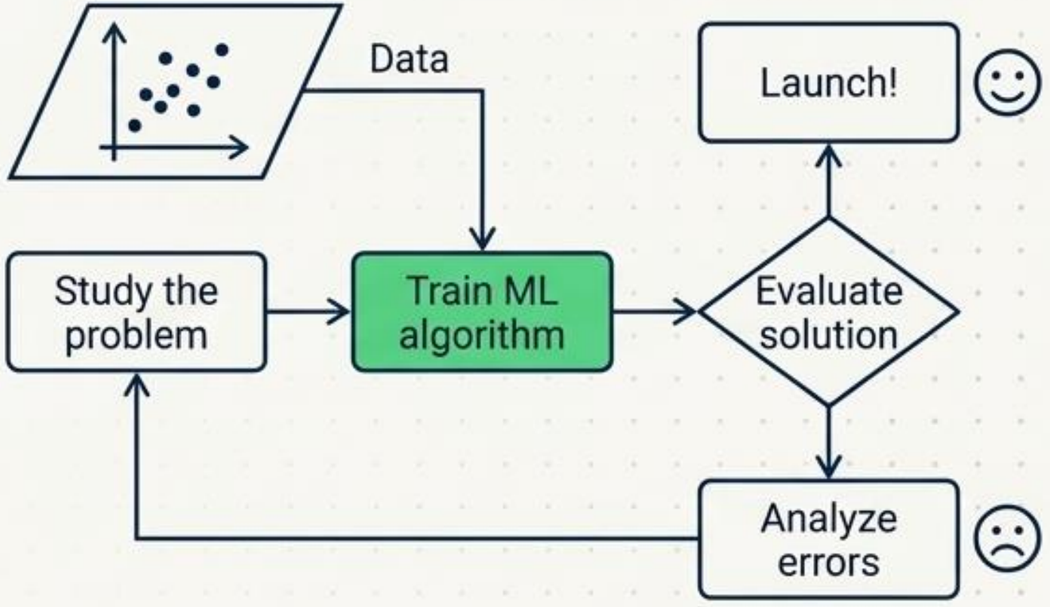
The Traditional Approach

- Spammers adapt, requiring humans to keep writing endless new If/Then rules forever.



The ML Approach

- The system automatically learns which words are good predictors by detecting unusually frequent patterns.



ML replaces endless rule-writing with automated pattern recognition.

Traditional Software Rules Fail When Problems Adapt Dynamically

The Core Problem: Building a Traditional Spam Filter

In traditional programming, humans analyze spam emails, **identify patterns (e.g., '4U', 'Free Money')**, and hand-craft specific code rules. When spammers adapt, developers must manually write more rules forever.

1. Study Problem

Analyze spam emails & notice common suspicious words.

2. Write Rules

Code explicit IF-THEN conditions for each keyword.

3. Evaluate

Test filter accuracy on incoming inbox emails.

4. Spammers Adapt!

Spammers bypass rules -> Filter breaks -> Repeat cycle!

Interactive Question for Students

If spammers change spelling from 'FREE' to 'F R E E' or 'F_R_E_E', how many manual rules would your software team need to write and maintain every year?

Today’s Learning Journey

From the big idea to the main challenges

1. What is ML?
2. Why ML?
3. Applications
4. Types of ML
5. Challenges

Goal: understand WHAT each concept is, WHY it matters, and WHEN to use it.

What Exactly IS Machine Learning?

1997 - Tom Mitchell

"A computer program is said to learn from experience **E** with respect to some task **T** and some performance measure **P**, if its performance on **T**, as measured by **P**, improves with experience **E**."

E **T** **P**

1959 - Arthur Samuel

"The field of study that gives computers the ability to learn without being explicitly programmed."



E = Training Data

e.g., examples of spam/ham emails



T = The Task

e.g., flagging spam in a user's inbox



P = Accuracy

e.g., ratio of correctly flagged emails

Machine Learning Automates Rule Creation directly from Examples

The Paradigm Shift

Instead of hardcoding rules manually, **we feed historical data and expected outputs into a Machine Learning algorithm.**

The computer automatically learns patterns, word frequencies, and statistics to generate its own decision model!

Data + Expected Answers -> ML Training -> Learned Model

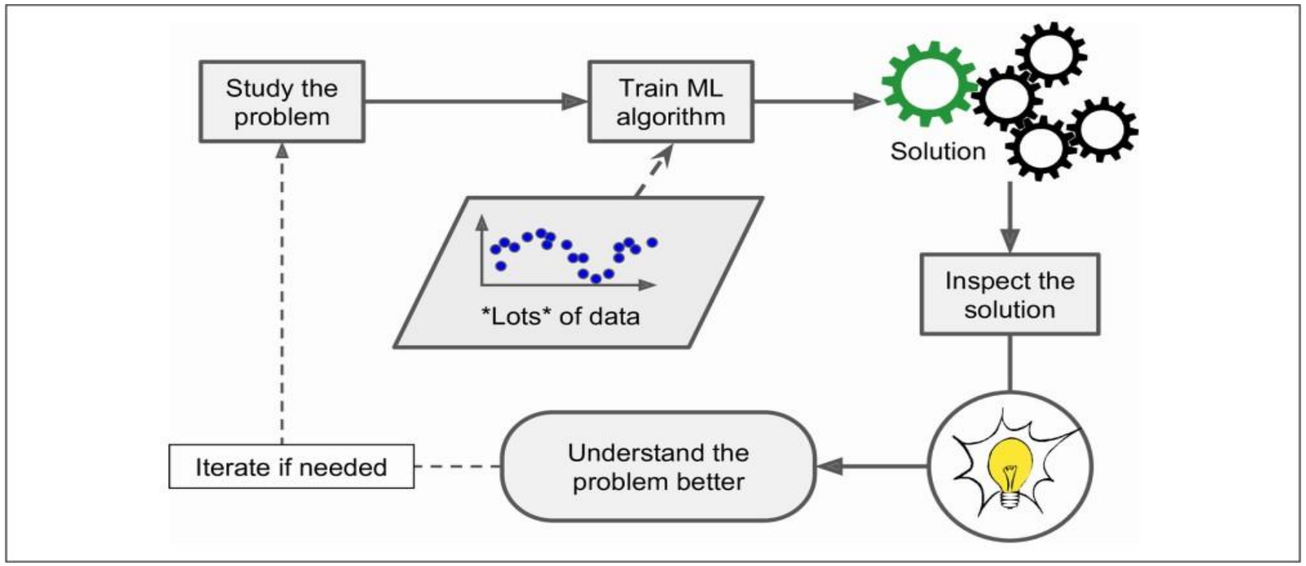
Academic Definitions

1. General Definition:
The science and art of programming computers so they can learn from data.

2. Arthur Samuel (1959):
The field of study that gives computers the ability to learn without being explicitly programmed.

3. Tom Mitchell (1997 - Engineering Definition):
A computer program is said to learn from experience E with respect to task T and performance measure P, if its performance on T, as measured by P, improves with experience E.

ML Can Help Humans Learn



- Applying ML techniques to dig into large amounts of data can help discover patterns that were not immediately apparent. This is called data mining.

When is Machine Learning the Right Tool?

Real Problems. Real Examples. Real Impact.

ML shines when rules are not enough!

1. Spam Email Detection

Traditional rules may fail because spam emails keep changing. ML learns from examples.

SPAM Win a FREE iPhone now!!!

SPAM Limited offer! Click here!

HAM Your meeting schedule for tomorrow

Train



Learns patterns from many emails

2. Image Recognition

Writing rules for every object is impossible. ML learns from thousands of images.



Train



Learns features automatically

3. Fraud Detection

Fraud patterns are complex and constantly changing. ML detects unusual transactions.

$20	Supermarket	Legitimate
$1,200	Flight Ticket	Legitimate
$5,000	Electronics	Suspicious
$8,000	Luxury Store	Fraud

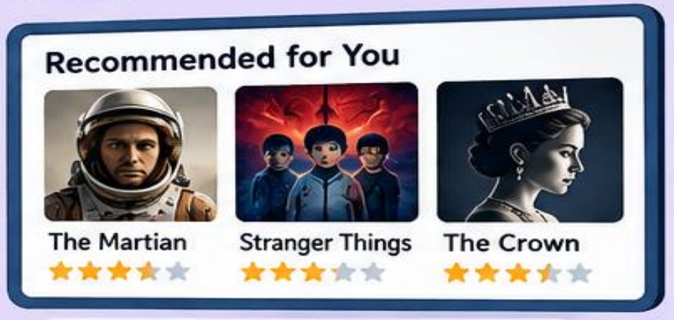


Spots abnormal behavior

4. Recommendation Systems

User preferences are complex. ML learns from past behavior to make personalized recommendations.





Learns from your previous choices

Machine Learning is the right tool when the problem is **too complex** for hand-written rules, the data is **large or changing**, and we want the system to **learn and improve** over time.

When is ML the Right Tool?



Long Rules

Replaces traditional solutions that require endless fine-tuning and long lists of brittle rules.



No Traditional Solution

Solves highly complex problems where traditional programming logic fails entirely.



Fluctuating Environments

Adapts automatically to new data as environments and user behaviors change.



Data Mining

Digs into massive datasets to discover hidden patterns and help humans learn.

Real-World Applications of Machine Learning

1 Product Inspection on a Production Line

Analyzing images of products to automatically classify them (e.g., good vs. defective).

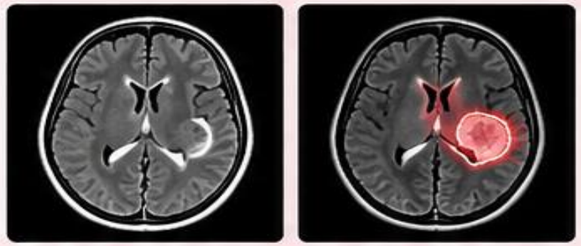


Task: Image classification

Typical models: CNNs

2 Tumor Detection in Brain Scans

Detecting tumors in brain scans and identifying the affected regions.



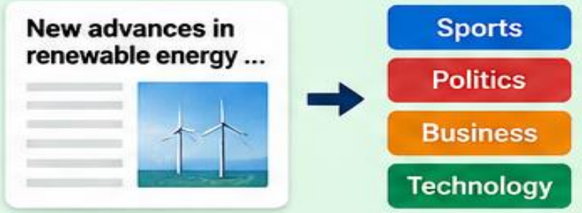
Original Scan | Detected Tumor

Task: Semantic segmentation

Typical models: CNNs

3 News Article Classification

Automatically classifying news articles into categories (e.g., sports, politics, business, technology).



Task: Natural language processing (NLP) – text classification

Typical models: RNNs, CNNs, or Transformers

4 Detecting Offensive Comments

Automatically flagging offensive or harmful comments on discussion forums.

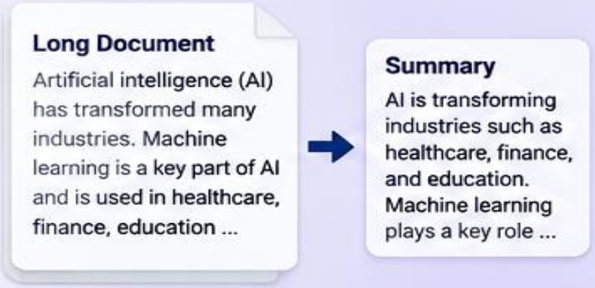


Task: Text classification

Typical models: RNNs, CNNs, or Transformers

5 Text Summarization

Summarizing long documents automatically.



Task: Text summarization

Typical models: RNNs, CNNs, or Transformers

6 Chatbots & Personal Assistants

Creating a chatbot or personal assistant that understands language and answers questions.

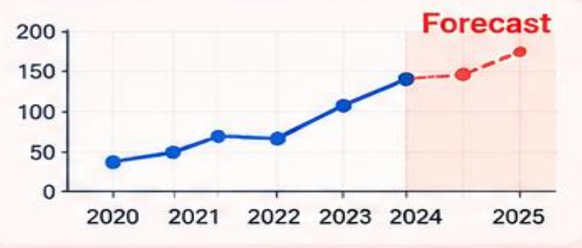


Task: Natural language understanding (NLU) and question-answering

Typical models: RNNs, CNNs, or Transformers

7 Revenue Forecasting

Forecasting a company's revenue next year based on historical data and performance metrics.



Task: Regression

Typical models: Linear Regression, Polynomial Regression, Regression SVM, Random Forest, ANN, RNNs, CNNs, or Transformers

8 Other Real-World Examples

- **Healthcare** (disease prediction)
- **Autonomous** vehicles
- **Recommendation systems**
- **Fraud detection** (in finance)

Machine Learning is everywhere!

More Real-World Applications of Machine Learning

1 Speech Recognition

Making your app react to voice commands.



Task: Speech recognition
Typical models: RNNs, CNNs, or Transformers

2 Credit Card Fraud Detection

Detecting fraudulent transactions.



$12	Coffee Shop	Legitimate
$250	Electronics	Legitimate
$1,200	Unknown Store	Suspicious
$3,500	Overseas	Fraud

Task: Anomaly detection
Typical models: Isolation Forest, Autoencoders, or other anomaly detection techniques

3 Customer Segmentation

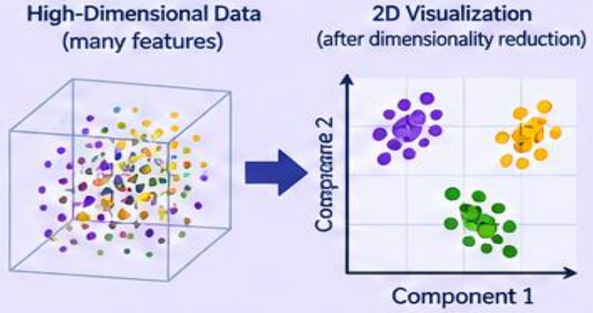
Grouping clients based on their purchases to design a different marketing strategy for each segment.



Task: Clustering
Typical models: K-Means, Hierarchical Clustering, DBSCAN, etc.

4 Data Visualization

Representing a complex, high-dimensional dataset in a clear and insightful diagram.



Task: Dimensionality reduction
Typical models: PCA, t-SNE, UMAP, etc.

5 Product Recommendation

Recommending a product that a client may be interested in, based on past purchases.



Task: Recommender system
Typical models: Artificial Neural Networks (ANN), Collaborative Filtering, etc.

6 Game Playing (Intelligent Bot)

Building an intelligent bot for a game using reinforcement learning.



Task: Reinforcement Learning (RL)
Typical models: Q-Learning, Deep Q-Network (DQN), Policy Gradient, etc.

Key Takeaway

Machine Learning is everywhere!

From speech and images to recommendations and intelligent systems, ML helps us make sense of data and solve real-world problems.

Hear · See · Read · Recommend · Play

Three Perspectives to Classify Machine Learning Systems

We can look at a machine learning system from three different perspectives:

1 Are they trained with human supervision?

Does the training data have the correct answers (labels) provided by humans?

Supervised Learning

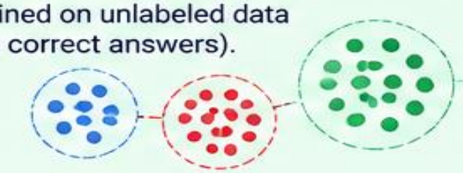
Trained on labeled data (with correct answers).



Example: Image classification (cat vs dog)

Unsupervised Learning

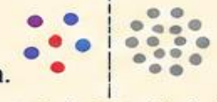
Trained on unlabeled data (no correct answers).



Example: Customer segmentation (clustering)

Semi-Supervised Learning

Uses a small amount of labeled data and a large amount of unlabeled data.



Example: Image classification with few labels

Reinforcement Learning

Learns by interacting with an environment and receiving rewards or penalties.



Example: Game playing, robot control

2 Can they learn incrementally on the fly?

Do they learn once using all the data, or can they keep learning from new data?

Batch Learning

Trained once using the entire dataset. If new data arrives, the model usually needs to be retrained.



Example: Training a model on a fixed dataset of images.

Online Learning

Learns incrementally from new data as it arrives (one sample or small batches at a time).



Example: Spam filter that keeps learning from new emails.

3 How do they generalize to new data?

How do they make predictions for examples they have not seen before?

Instance-Based Learning

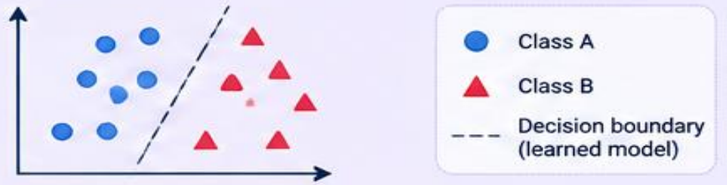
Compares the new example with similar training examples and uses them to predict the output.



Example: k-Nearest Neighbors (KNN).

Model-Based Learning

Learns a general model or pattern from the data (e.g., a mathematical function) and uses it to predict new outputs.



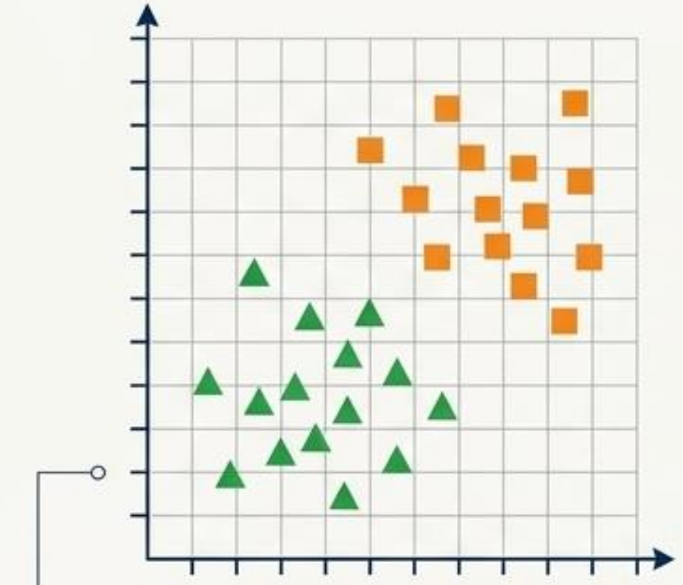
Example: Linear Regression, Decision Trees, Neural Networks, SVM, etc.

Key Takeaway: The same machine learning system can be described from all three perspectives, e.g., a spam filter can be supervised, online, and model-based.

Different views...
A clearer understanding!

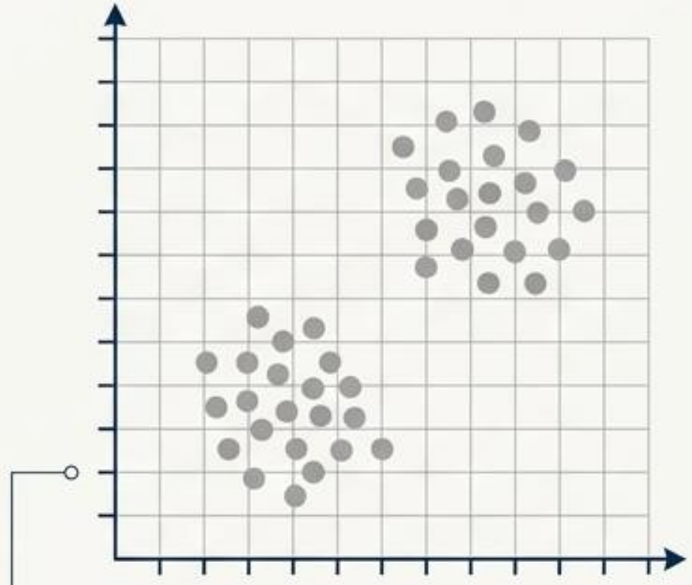
Pathway 1: Supervision Level (Supervised vs. Unsupervised)

Supervised Learning



The training set you feed to the algorithm includes the desired solutions (labels).

Unsupervised Learning



The training data is unlabeled. The system tries to learn the underlying structure without a teacher.

Supervised Learning: Classification vs. Regression Problems

Definition of Supervised Learning

The training dataset fed to the learning algorithm includes the target ground-truth answers, called LABELS.

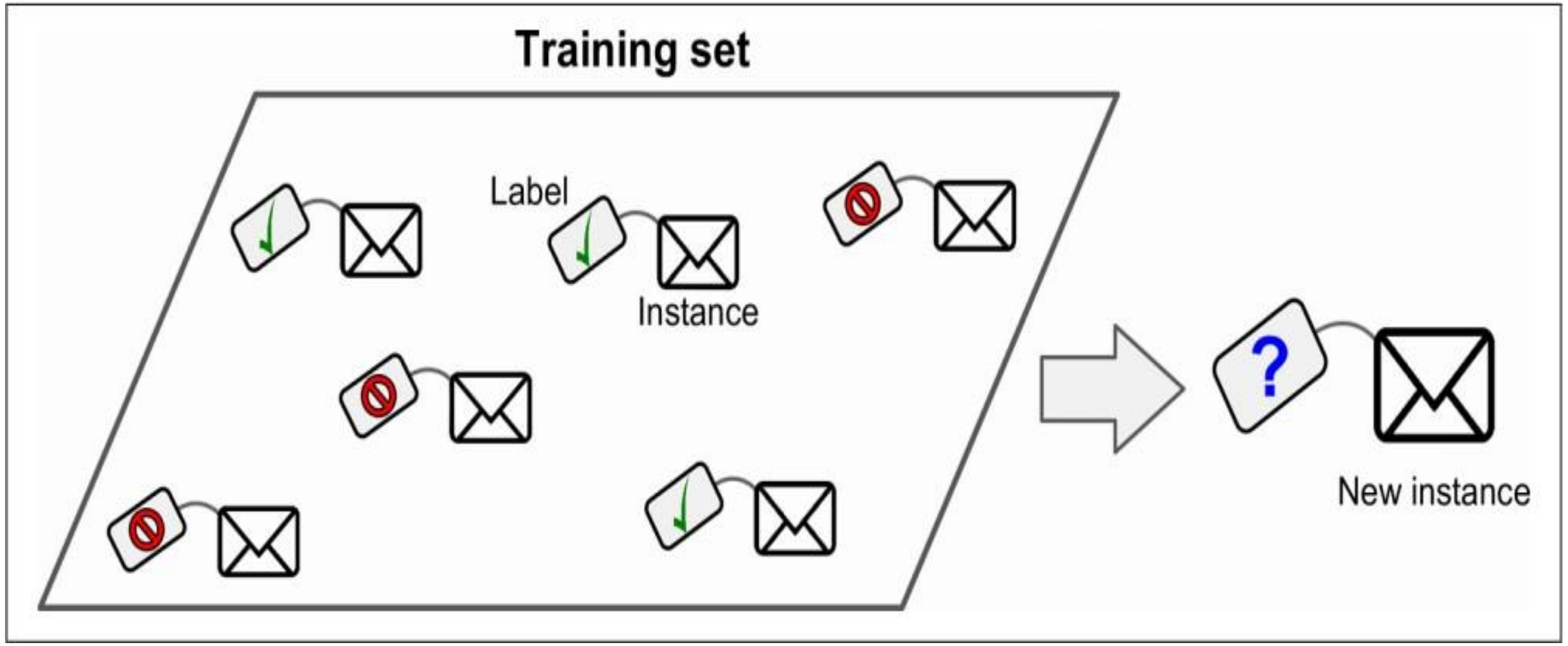
Dimension	Classification	Regression
Target Output	Discrete class label (e.g., Category A vs Category B).	Continuous numerical value (e.g., Price, Temperature).
Example Application	Spam vs. Ham email filter, Offensive comment detector.	Predicting housing prices based on GDP/income features.
Key Algorithms	Logistic Regression, SVMs, k-NN, Decision Trees.	Linear Regression, Polynomial Regression, Neural Nets.

Important Supervised Learning Algorithms

• k-Nearest Neighbors (k-NN) • Linear Regression • Logistic Regression • Support Vector Machines (SVMs) • Decision Trees & Random Forests • Neural Networks

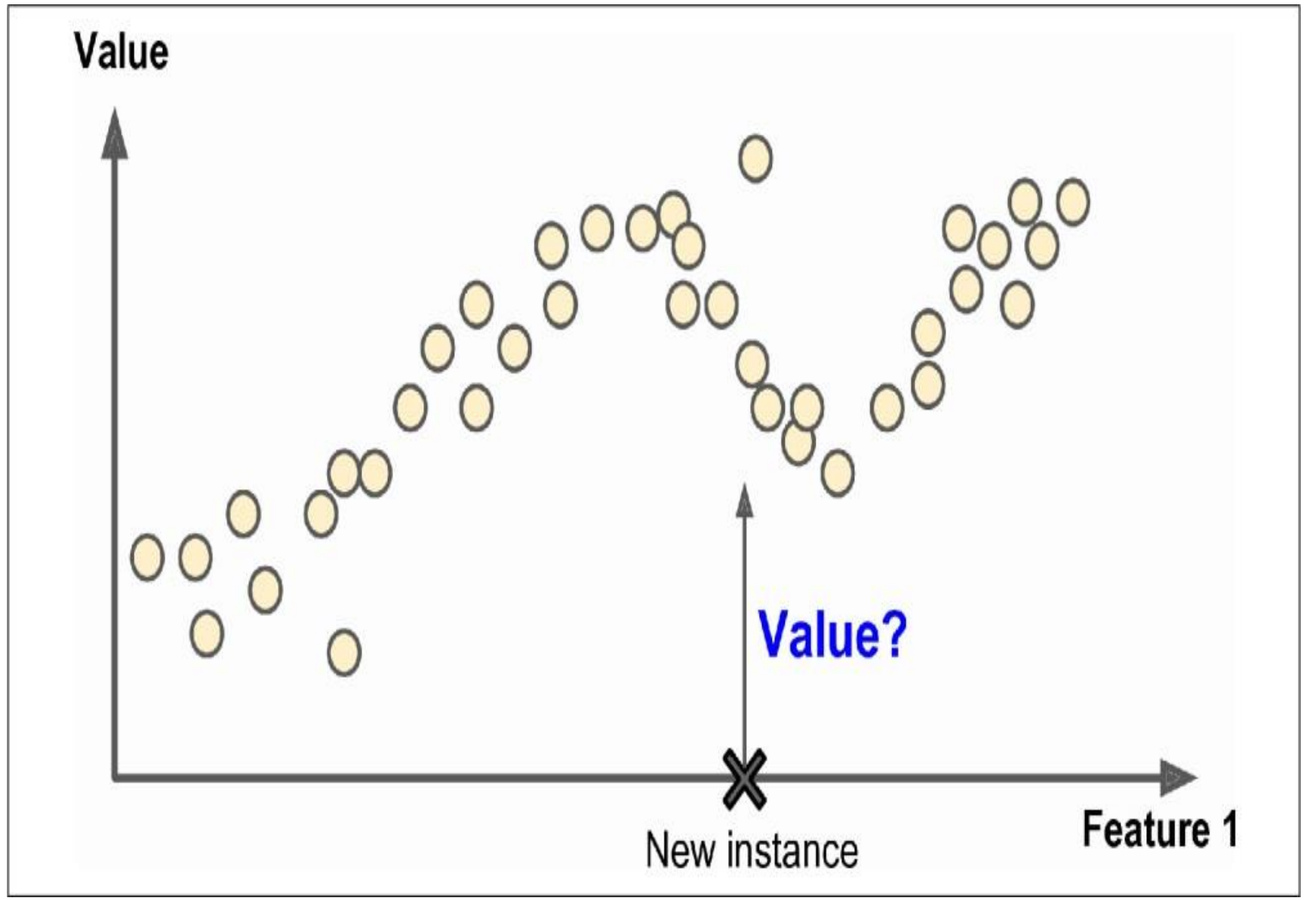
1. Supervised Learning: Classification

A labeled training set for spam classification



1. Supervised Learning: Regression

Predict a value, given an input feature (there are usually multiple input features, and sometimes multiple output values)

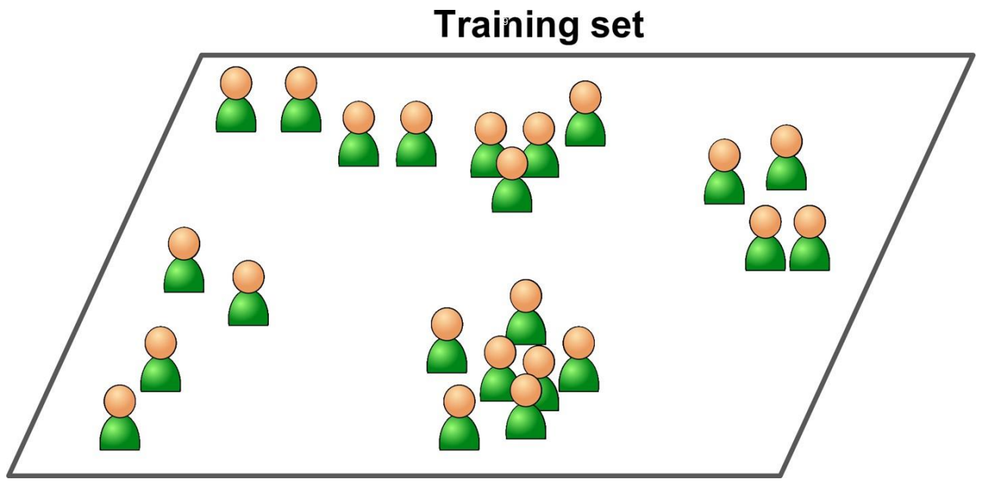


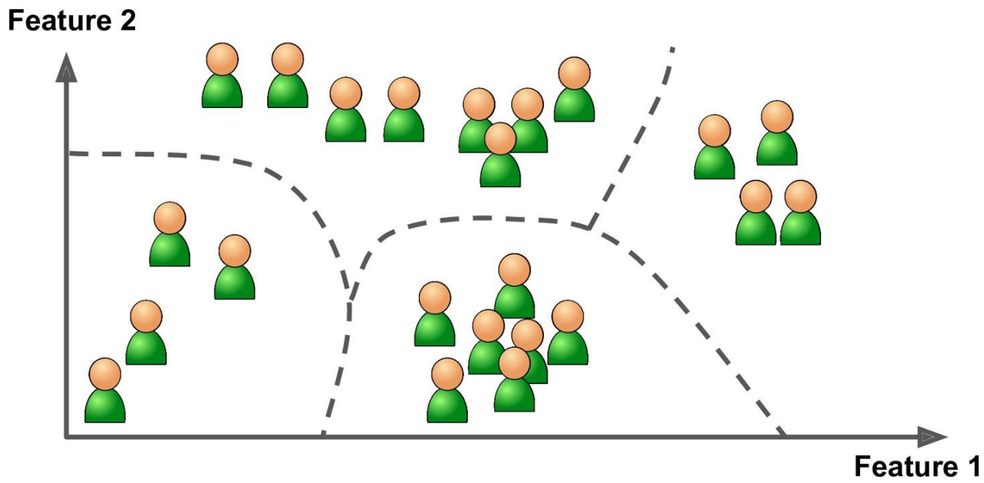
The Most Important Supervised Learning Algorithms

- k-Nearest Neighbors.
- Linear Regression.
- Logistic Regression.
- Support Vector Machines (SVMs).
- Decision Trees and Random Forests.
- Neural networks.

2. Unsupervised Learning

Clustering





The Unsupervised Learning Toolkit

Finding hidden patterns in data without labeled answers

Let the data speak!

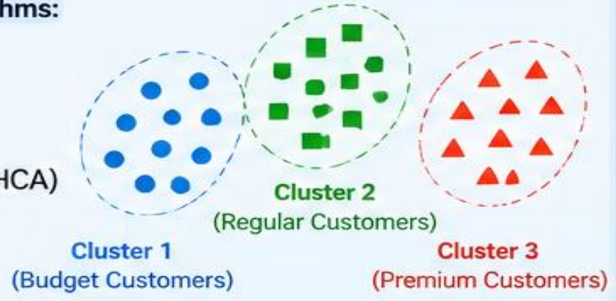
1 Clustering

Goal: Group similar data together

Finds natural groups in data based on similarity.

Common Algorithms:

- K-Means
- DBSCAN
- Hierarchical Clustering (HCA)



Real-World Example:

Customer Segmentation

Grouping customers based on their purchases so each group gets a tailored marketing strategy.

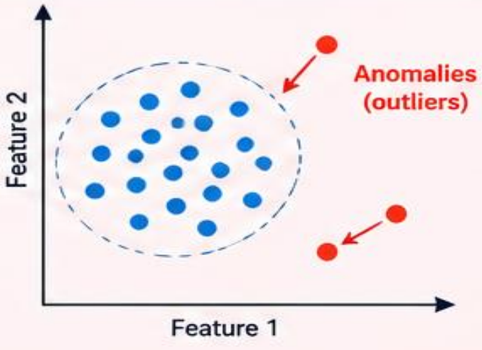
2 Anomaly & Novelty Detection

Goal: Find rare or unusual items

Identifies data points that are very different from the rest.

Common Algorithms:

- One-class SVM
- Isolation Forest
- Local Outlier Factor (LOF)



Real-World Example:

Credit Card Fraud Detection

Detecting unusual transactions that are different from a user's normal behavior.

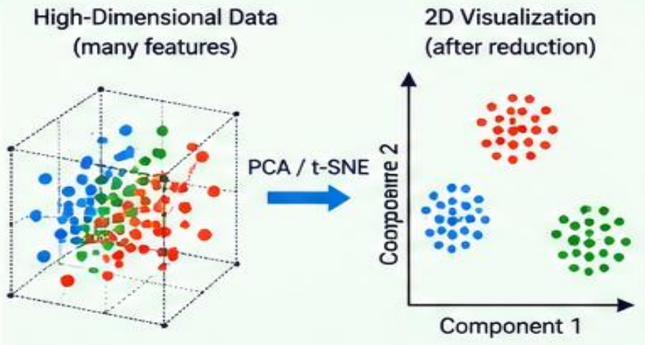
3 Visualization & Dimensionality Reduction

Goal: Simplify high-dimensional data

Reduces the number of features while keeping the important structure.

Common Algorithms:

- PCA
- Kernel PCA
- LLE
- t-SNE
- UMAP



Real-World Example:



Visualizing Image Data

Reducing image features to 2D to visualize groups of similar images (e.g., handwritten digits or faces).

4 Association Rule Learning

Goal: Find interesting relationships

Discovers patterns and relationships between variables in data.

Common Algorithms:

- Apriori
- Eclat



If a customer buys **bread** → they also buy **milk** (support = 30%, confidence = 70%)

If a customer buys **diapers** → they also buy **baby wipes** (support = 20%, confidence = 60%)

Real-World Example:

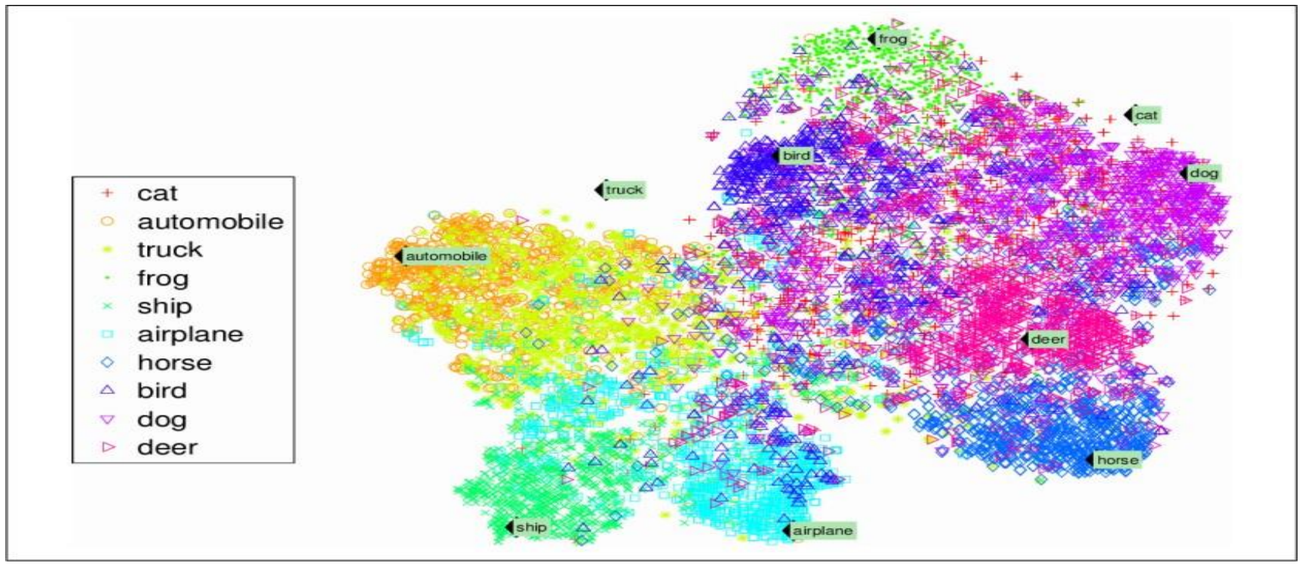
Market Basket Analysis

Finding products that are often bought together (e.g., bread and milk) to improve store layout and recommendations.

Key Takeaway: Unsupervised learning helps us discover hidden structure, rare events, and meaningful patterns in unlabeled data, which is **everywhere** in the real world!

No labels? No problem!

Example: t-SNE visualization highlighting semantic clusters



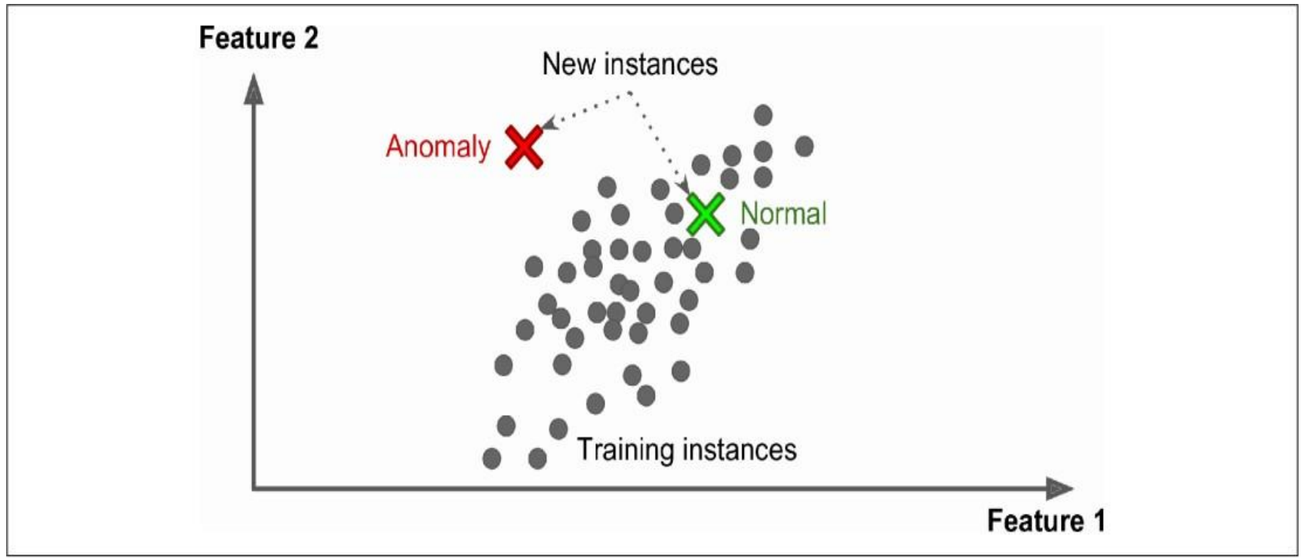
- Visualization algorithms: The user feeds them a lot of complex and unlabeled data, and they **output a 2D or 3D representation** of the data that can easily be plotted.

Other Unsupervised Learning

- Dimensionality Reduction: The goal is to **simplify the data without losing too much information.** One way to do this is to **merge several correlated features** into one.

- Association Rule Learning: It **dig into large amounts of data** and discover **interesting relations between attributes.**

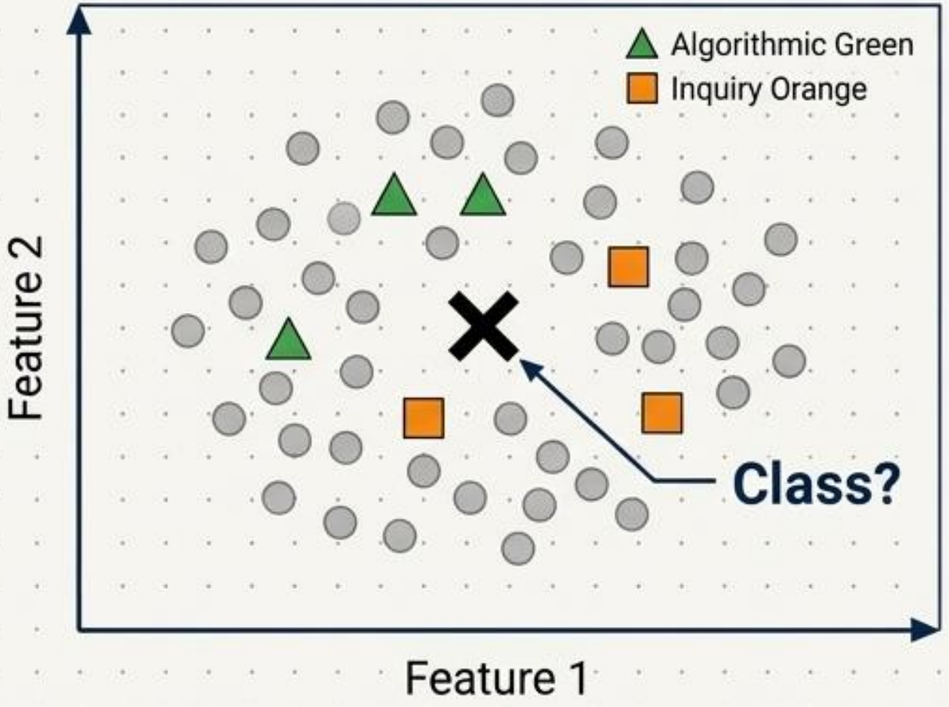
Anomaly Detection



- The system is **shown mostly normal instances during training**, so it **learns to recognize them**; then, when it sees a new instance, it can tell whether it looks like a normal one or whether it is likely an anomaly. Similar to novelty detection.

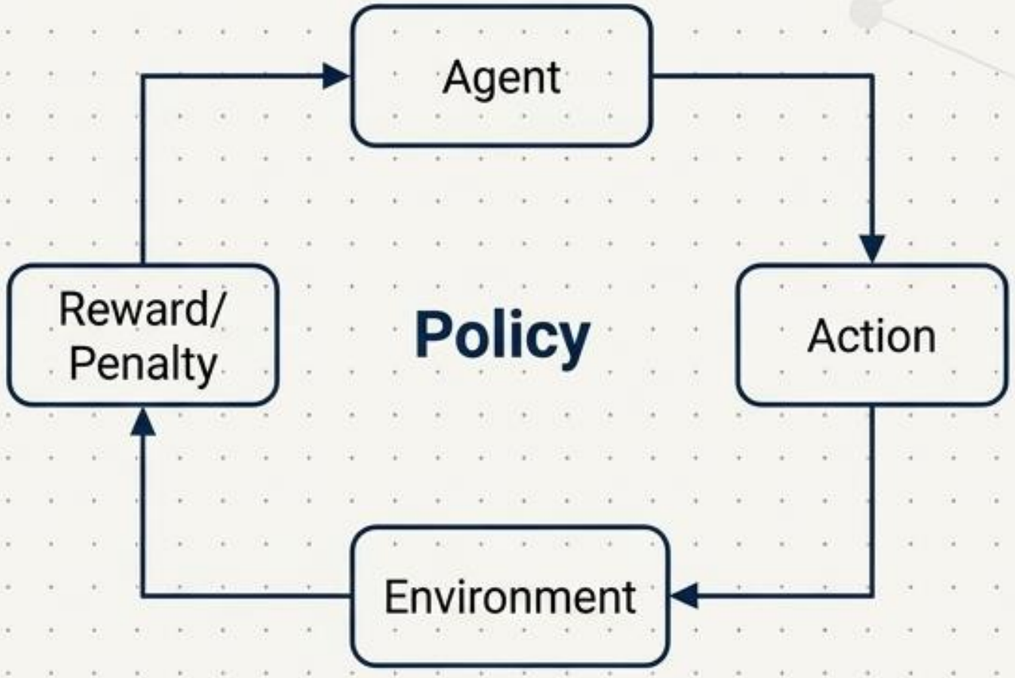
Completing the Spectrum: Semi-Supervised & RL

Semi-Supervised Learning



Algorithms that deal with partially labeled training data (a few labeled, many unlabeled).

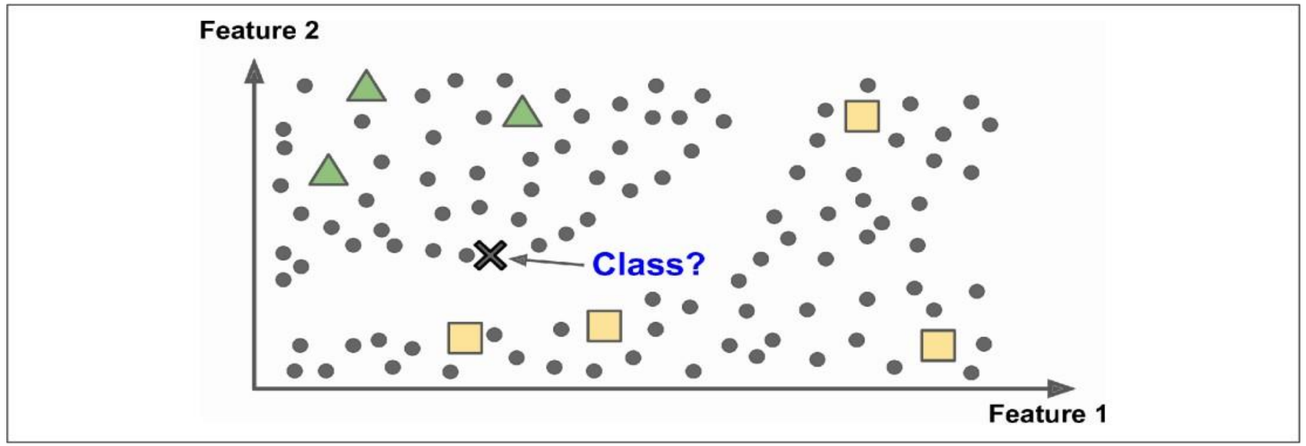
Reinforcement Learning (RL)



An Agent observes an environment, acts, and gets Rewards or Penalties. It learns by itself the best strategy, called a Policy. (e.g., AlphaGo, walking robots).

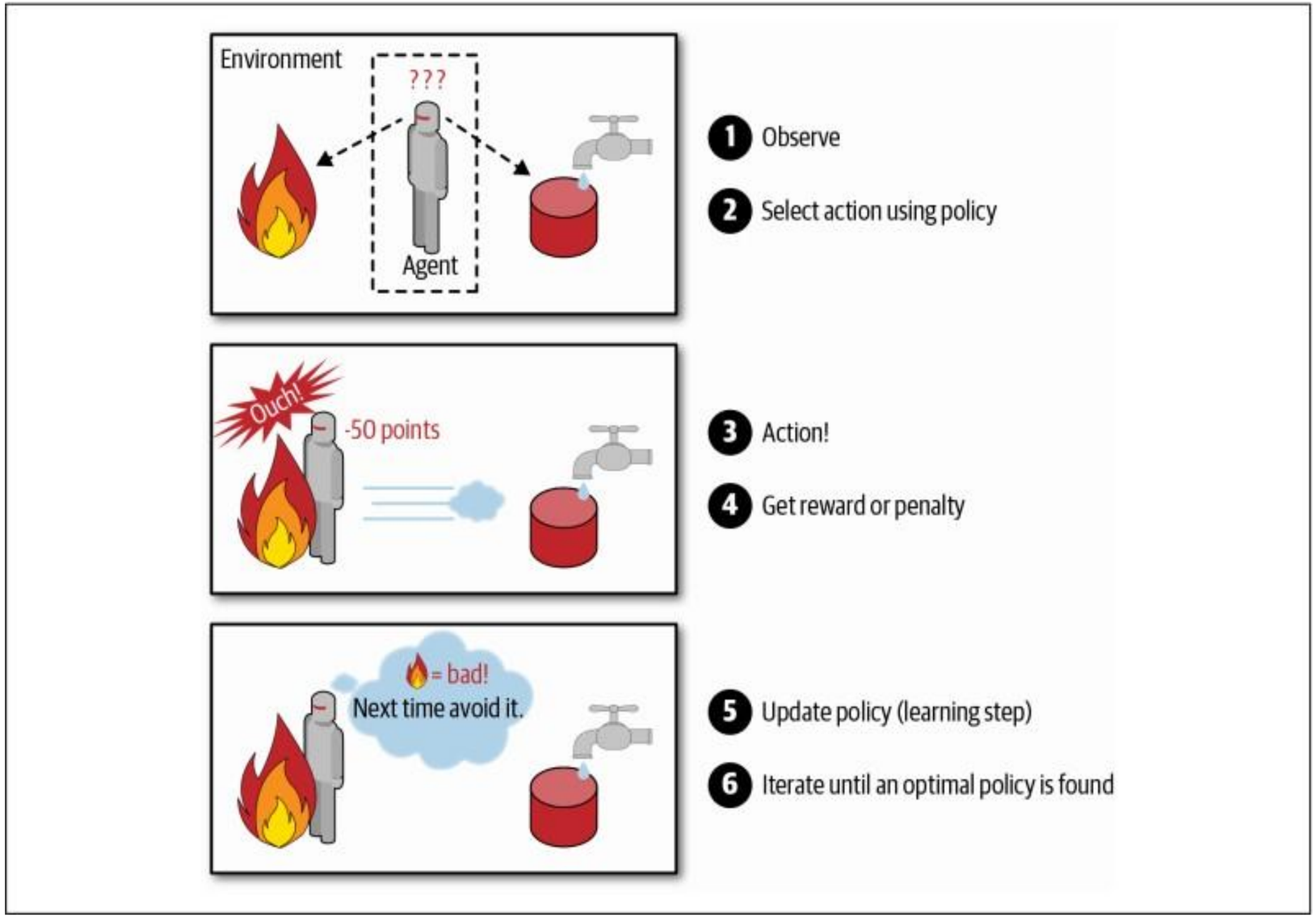
3. Semi-Supervised Learning

Semi-supervised learning with two classes (triangles and squares): the unlabeled examples (circles) help classify a new instance



- For example, Google Photos.
- Most semi-supervised learning algorithms are combinations of unsupervised and supervised algorithms.

4. Reinforcement Learning



4. Reinforcement Learning (cont'd)

The leaning system is called an agent.

It must then learn by itself what is the **best strategy**, called a policy, to get the most reward over time.

A policy defines **what action the agent should choose** when it is in a given situation.

Many robots implement Reinforcement Learning algorithms to

learn how to walk.

AlphaGo was just applying the policy it had learned.

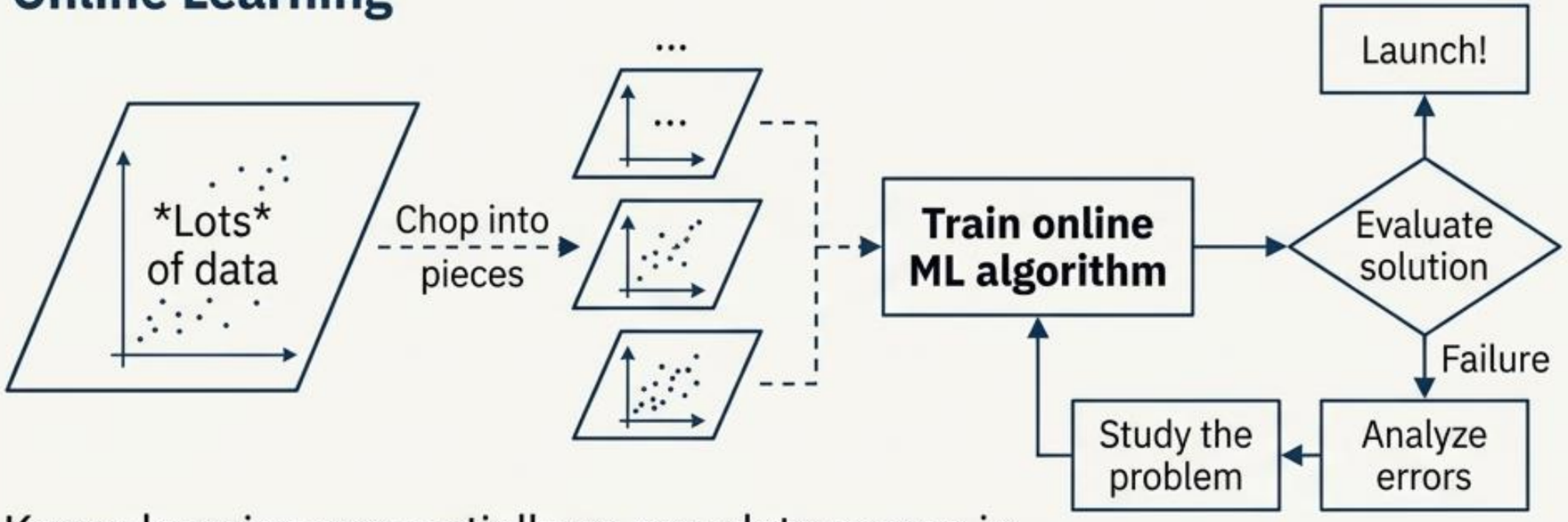
Pathway 2: Learning Over Time (Batch vs. Online)

Batch Learning



Learns from all available data at once.
Takes immense time/resources; cannot learn incrementally on the fly.

Online Learning



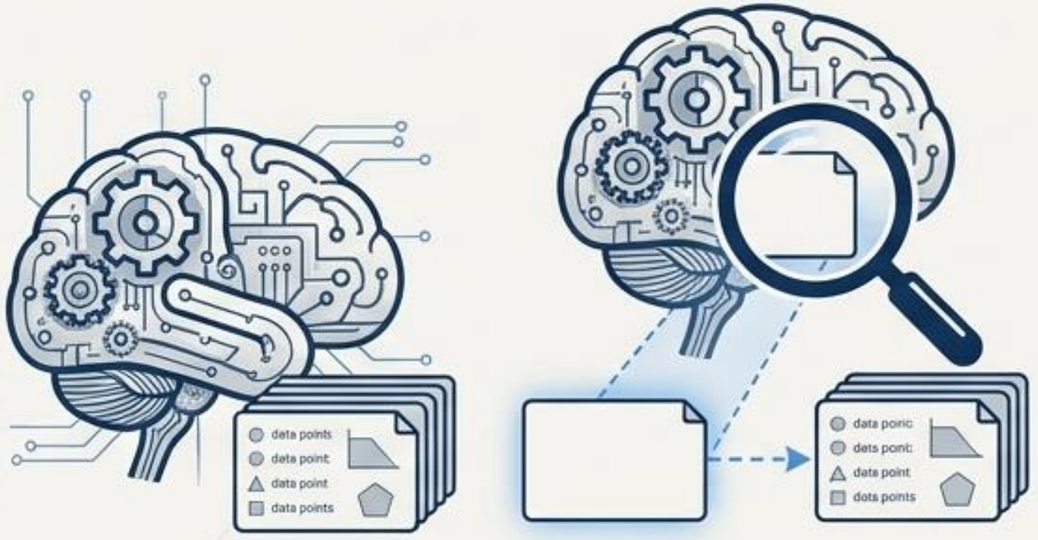
Keeps learning sequentially as new data comes in.

Out-of-core learning: Essential for huge datasets. Loads part of the data, trains, and repeats until the whole dataset is processed.

Which would you use for predicting daily stock market prices?

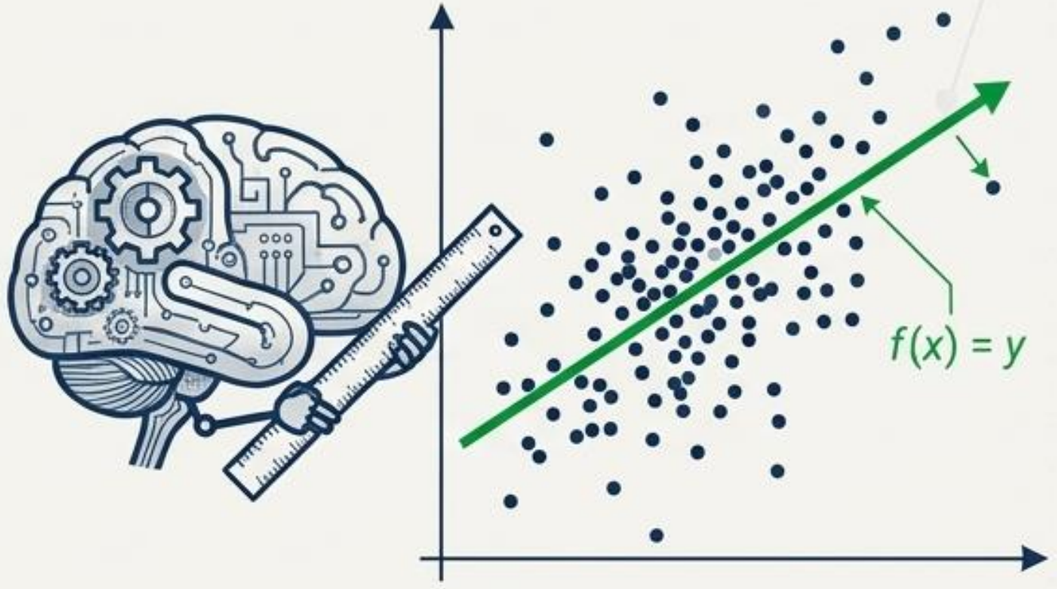
Pathway 3: How Do They Generalize?

Instance-Based Learning



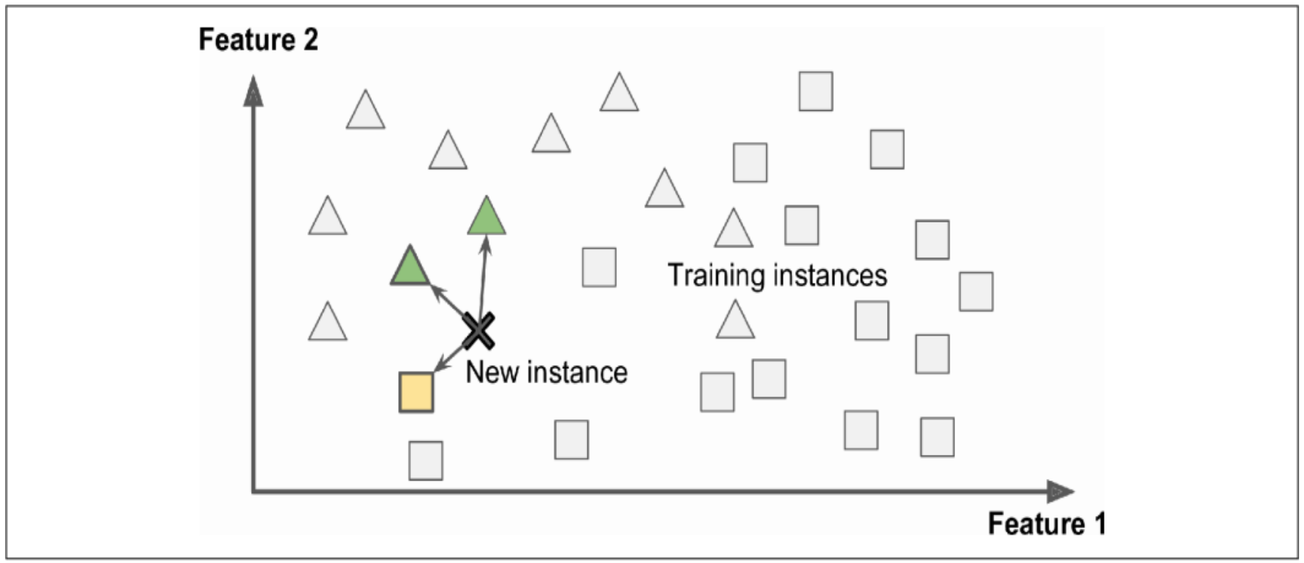
The system learns the examples by heart, then generalizes to new cases using a similarity measure (comparing new instances to memorized ones).

Model-Based Learning



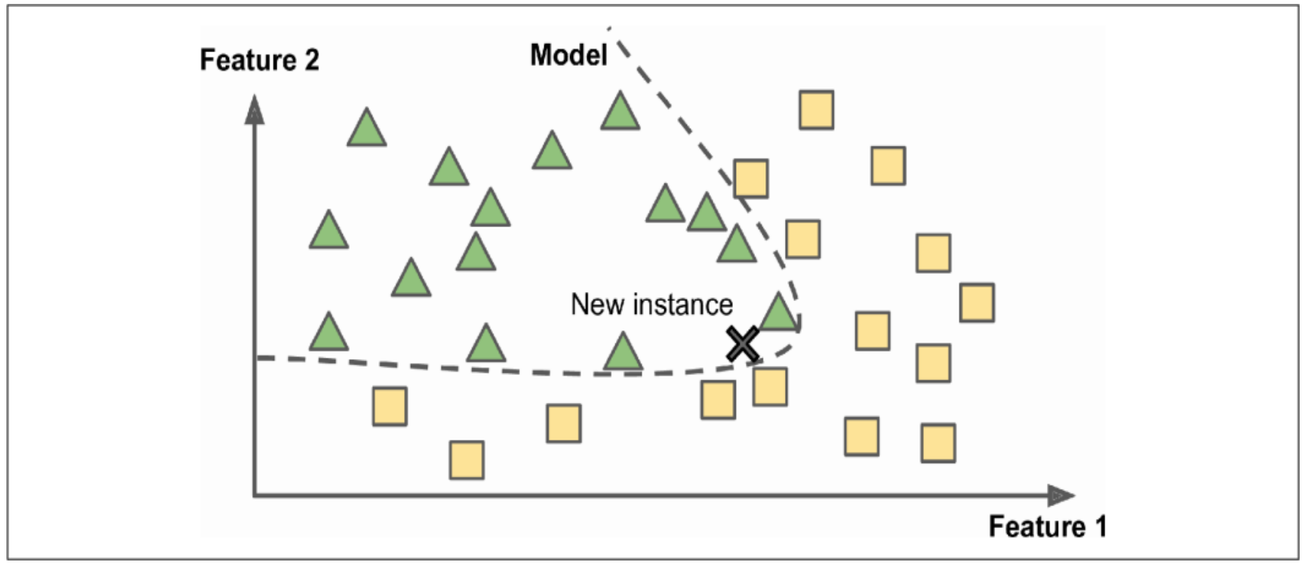
The system builds a mathematical model from a set of examples, then uses that underlying rule to make predictions on new data.

1. Instance-based learning



- The system learns the examples by heart, then **generalizes to new cases by using a similarity measure** to compare them to the learned examples.

2. Model-based learning



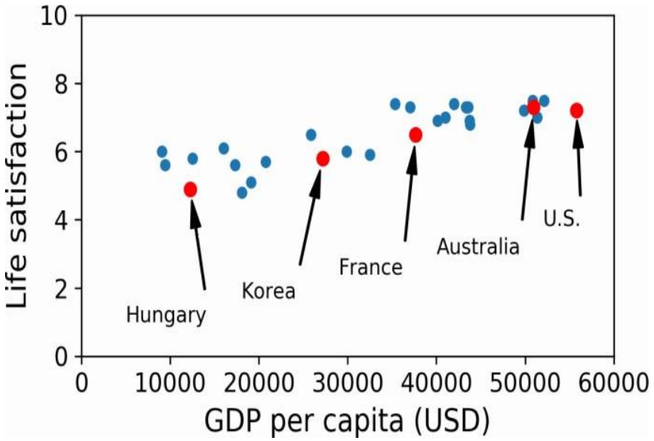
- Another way to generalize from a set of examples is to build a model of these examples and then use that model to make predictions.

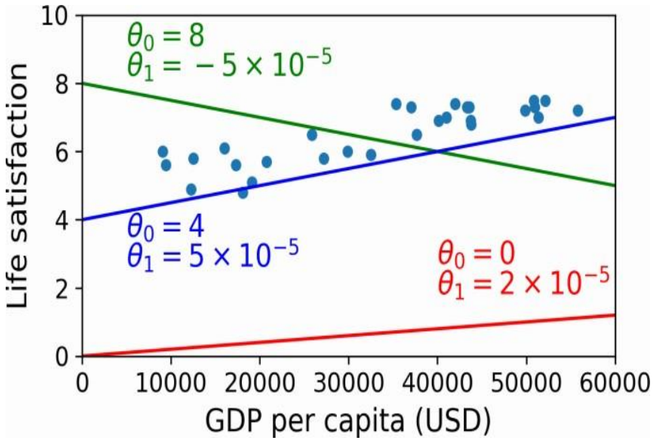
Example: Does Money Makes People Happier?

Based on gross domestic product (GDP) per capita

Country	GDP per capita (USD)	Life satisfaction
Hungary	12,240	4.9
Korea	27,195	5.8
France	37,675	6.5
Australia	50,962	7.3
United States	55,805	7.2

Model Selection





- life satisfaction$= \theta_0 + \theta_1 \times$ *GDP per capita*
- The user can either define a utility function (or fitness function) that measures **how good your model is**, or you can define a cost function that **measures how bad it is.**

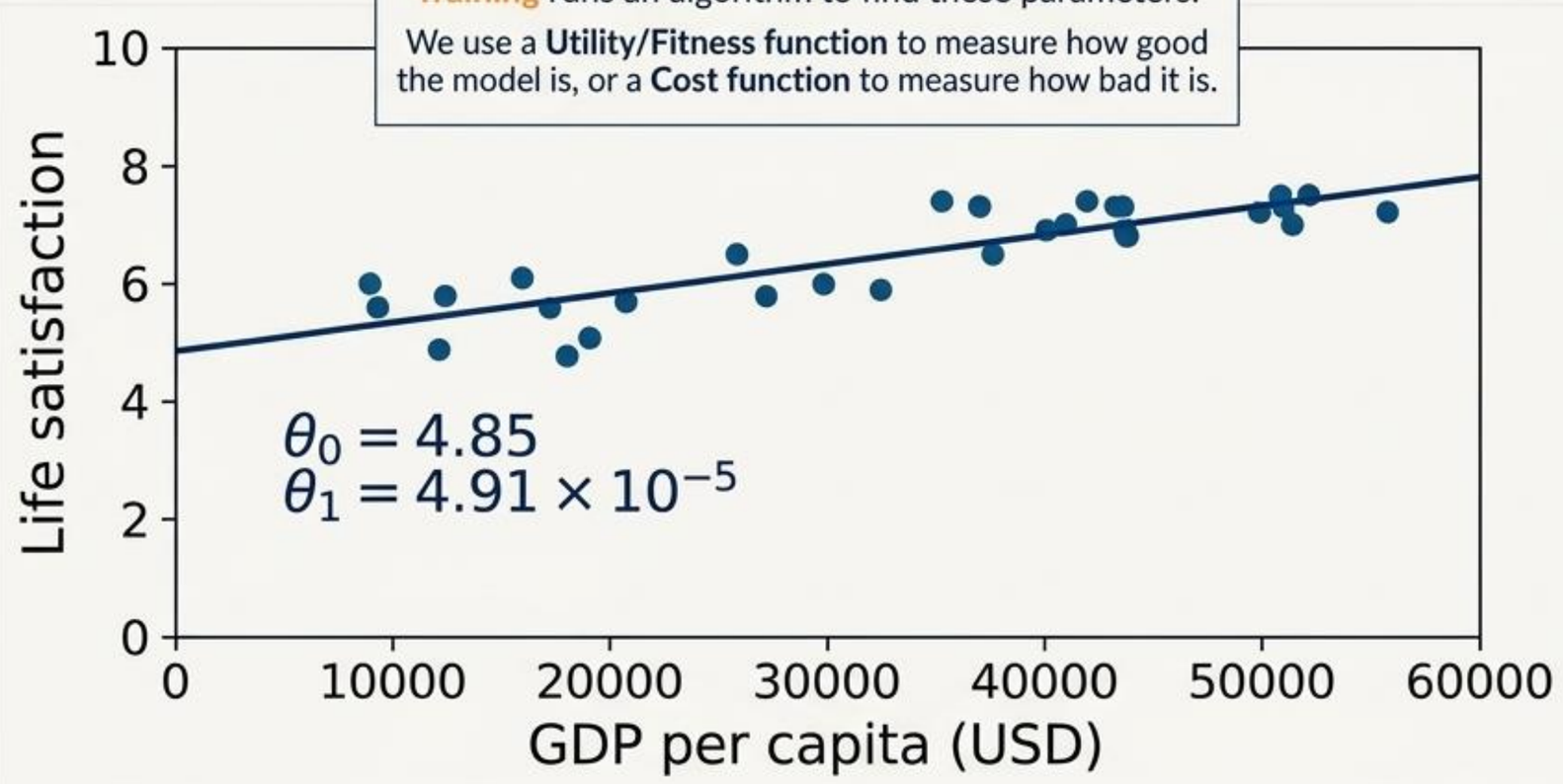
The Math Behind Model-Based Learning

$$\text{life_satisfaction} = \theta_0 + \theta_1 \times \text{GDP_per_capita}$$

θ_0 = **The Intercept**
(where the line starts)

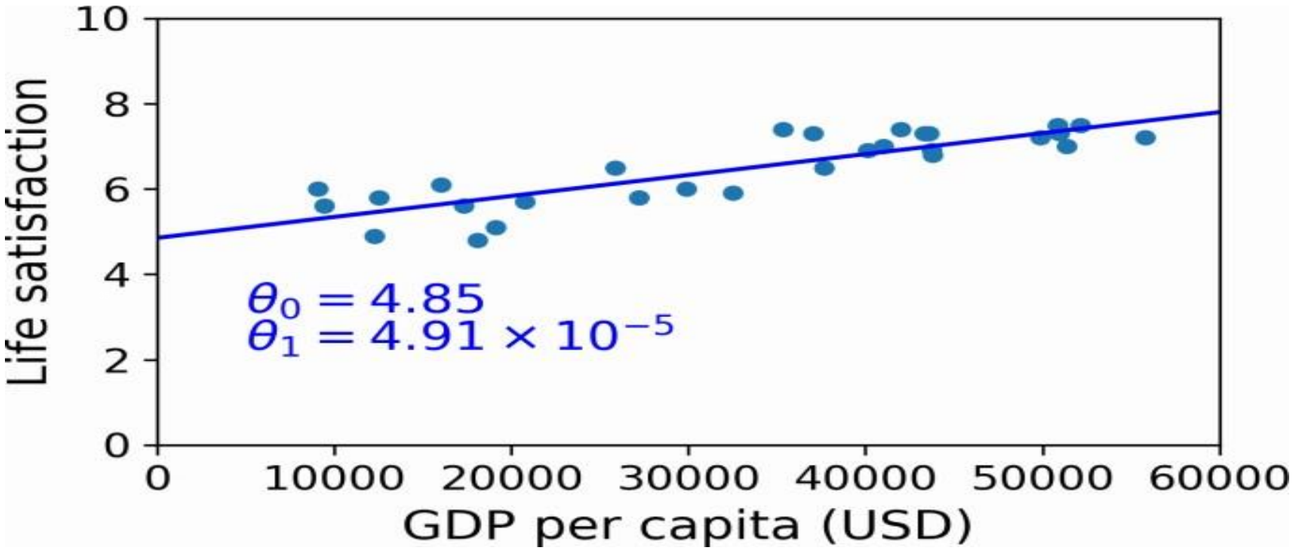
θ_1 = **The Slope**
(trajectory of the line)

Training runs an algorithm to find these parameters.
We use a **Utility/Fitness function** to measure how good the model is, or a **Cost function** to measure how bad it is.



Inference:
Applying the trained model to make predictions on new cases.

Using the Linear Model to Fit the Training Dataset



- Training a model means **running an algorithm to find the model parameters** that will make it **best fit the training data** (and hopefully make good predictions on new data).

Typical Project Workflow

1. The user studied the data.

2. The user selected a model.

3. The user trained it on the training data (i.e., the learning algorithm searched for the model parameter values that minimize a cost function).

4. Finally, the user applied the model to make predictions on new cases (this is called inference)



We have the definitions, the categories, and the math.

But what happens when it all goes wrong?

Academic Navy

Can you guess the two primary reasons a Machine Learning model fails in the real world?

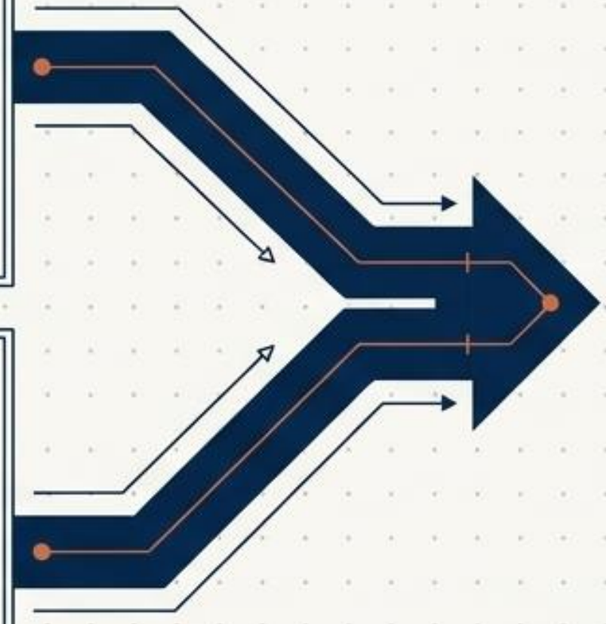
The Main Challenges of ML

Challenge 1: Bad Data

- Too little training data
- Non-representative data
- Poor quality data (errors/outliers)
- Irrelevant features

Challenge 2: Bad Algorithms

- Overfitting: The model memorizes noise instead of the underlying pattern.
- Underfitting: The model is too simple to learn the underlying structure.



Broken Model

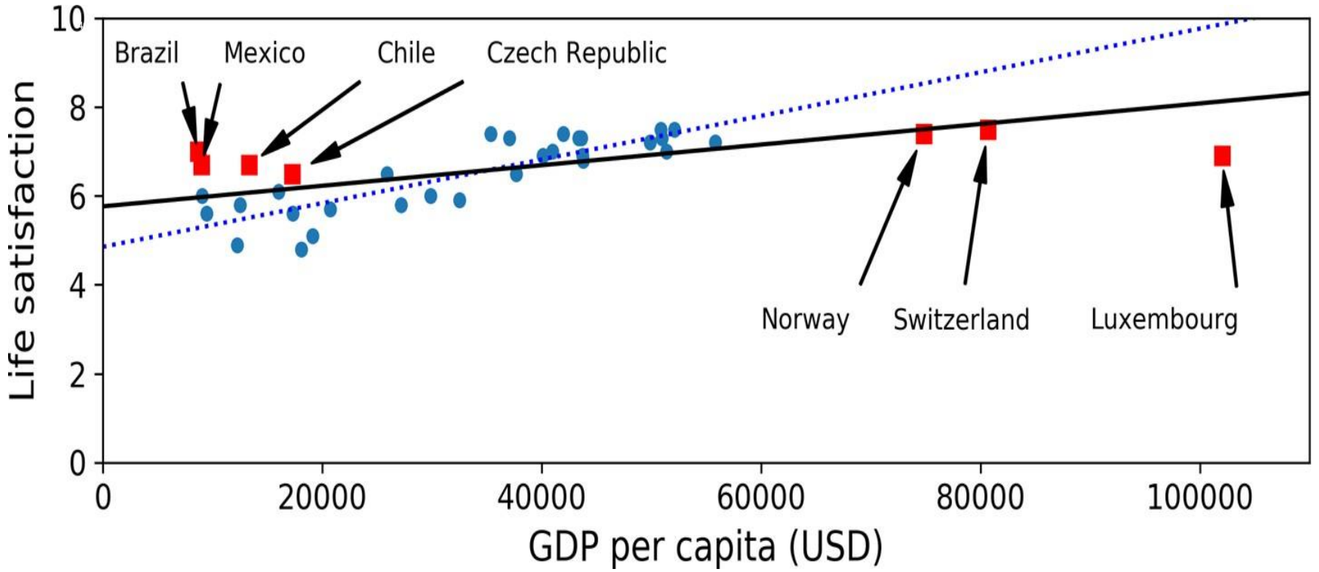
Garbage In = Garbage Out.
The system fails to generalize to new, real-world data.

1. Bad Data

- Insufficient Quantity of Training Data: It takes a lot of data for most ML algorithms to work properly.

- Nonrepresentative Training Data: If the **sample is too small**, you will have **sampling noise** (i.e., nonrepresentative data as a result of chance), but even **very large samples can be nonrepresentative** if the sampling method is flawed. This is called sampling bias.

A More Representative Training Sample



1. Bad Data (cont'd)

- Poor Quality Data: If your training data is full of **errors**, **outliers**, and **noise**, it will **make it harder for the system to detect the underlying patterns**, so your system is less likely to perform well.

- Irrelevant Features:Your system will only be capable of learning if the training data **contains enough relevant features and not too many irrelevant ones.**

Steps of Feature Engineering

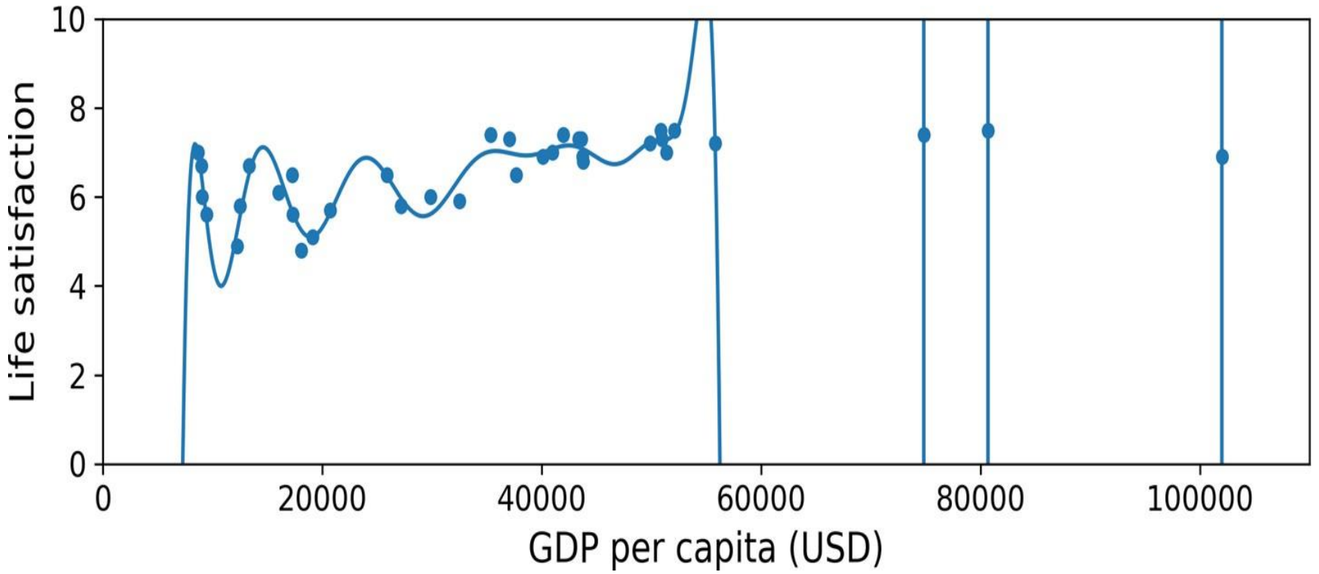
1. Feature Selection: Selecting the most useful features to train on among existing features.

2. Feature Extraction: Combining existing features to produce a more useful one - as we saw earlier, dimensionality reduction algorithms can help.

3. Creating new features by gathering new data.

2. Bad Algorithms

a. Overfitting the Training Data

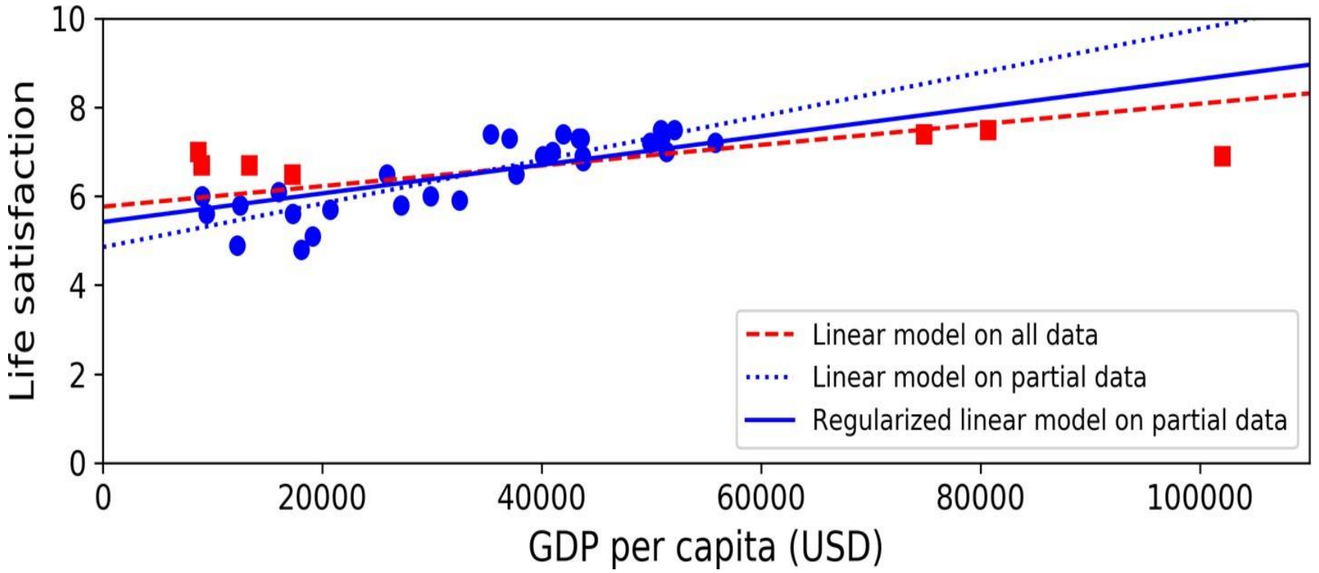


- It it means that the model performs well on the training data, but it **does not generalize well.**

Solutions for Overfitting

- Simplify the model by selecting one with **fewer parameters**, by reducing the number of attributes in the training data, or by constraining the model.
- Gather more training data.
- Reduce the noise in the training data (e.g., fix data errors and remove outliers).

Regularization Reduces the Risk of Overfitting



- Constraining a model to make it **simpler and reduce the risk of overfitting** is called regularization (hyperparameter). For example, the linear model we defined earlier has two parameters, θ_0 and θ_1 (**two degree of freedom**).

2. Bad Algorithms

b. Underfitting the Training Data

- It occurs when your model is too simple to learn the underlying structure of the data.

- Solutions for underfitting:
  - Select a more powerful model, with **more parameters.**
  - Feed better features to the learning algorithm (feature engineering).
  - Reduce the constraints on the model (e.g., reduce the regularization hyperparameter).


Machine Learning = A Brighter Future
Any Questions?
AI
Let's Learn Together!
Data
Algorithms
Better Tomorrow
Good Ideas Start Here

Key Takeaway: The ML Landscape Blueprint

